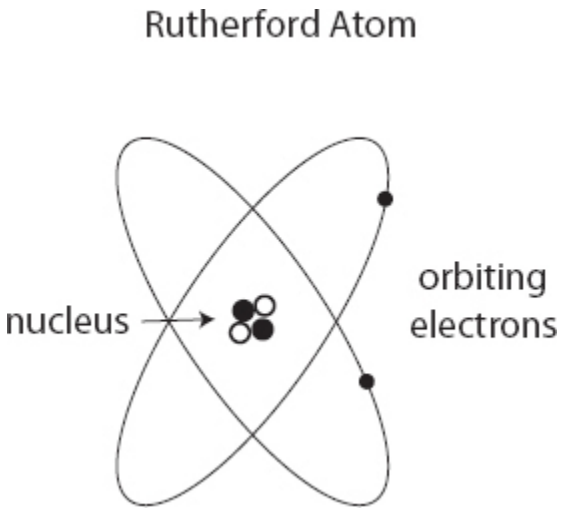


Chemically speaking, electrons are where it's at. Nuclei give atoms their heft, but outside of rare radioactive decays or fission/fusion reactions, they basically go along for the ride. The orbiting electrons, on the other hand, are light and jumpy, and their tendency to move around is what makes our lives interesting. Two or more atoms can share electrons, leading to chemical bonds. Under the right conditions, electrons can change their minds about which atoms they want to be associated with, which gives us chemical reactions. Electrons can even escape their atomic captivity altogether in order to move freely through a substance, a phenomenon we call "electricity." And when you shake an electron, it sets up a vibration in the electric and magnetic fields around it, leading to light and other forms of electromagnetic radiation.

To emphasize the idea of being truly point-like, rather than a small object but with some definite nonzero size, we sometimes distinguish between "elementary" particles, which define literal points in space, and "composite" particles that are really made of even smaller constituents. As far as anyone can tell, electrons are truly elementary particles. You can see why discussions of quantum mechanics are constantly referring to electrons when they reach for examples—they're the easiest fundamental particle to make and manipulate, and play a central role in the behavior of the matter of which we and our surroundings are made.